

pfSense Firewall Deployment Guide

Purpose

pfSense serves as the core routing, segmentation, and security enforcement device within the Virtual SOC environment. It is deployed between OpenStack's upstream router and the internal SOC network, acting as the control point for: - NAT - Firewall rule enforcement - DHCP services - SSH bastion access - WireGuard VPN - Suricata NIDS traffic visibility

Role in the Architecture

```
ext_net → ex-router → transit-net → pfSense WAN
                                   |
                                   └─ pfSense LAN → internal-net
(105.45.5.0/24)
```

pfSense must see all ingress, egress, and lateral traffic destined for internal hosts to ensure NIDS visibility and secure routing.

1. PFsense Instance Creation

OpenStack Steps

1. Create pfSense VM (m1.medium recommended).
2. Attach two ports:
3. **WAN** → transit-net (10.250.2.0/29)
4. **LAN** → internal-net (105.45.5.0/24)
5. Before booting, rename ports for clarity:

```
openstack port set --name pfw <wan-port-id>
openstack port set --name pfl <lan-port-id>
```

6. Replace LAN IP with the subnet gateway:

```
openstack port set --no-fixed-ip pfl
openstack port set --fixed-ip subnet=lan-test-subnet,ip-address=105.45.5.1
pfl
```

2. Interface Mapping

Inside pfSense shell (Option 8):

```
ifconfig vtnet0    # Identify MAC for WAN or LAN
ifconfig vtnet1
```

Map interfaces based on MAC and OpenStack port listing.

Assign: - **vtnet0** → **WAN** (10.250.2.5/29) - **vtnet1** → **LAN** (105.45.5.1/24)

Disable IPv6 on both.

3. LAN Configuration

- LAN IP: **105.45.5.1 /24**
 - DHCP enabled:
 - Range: 105.45.5.4 – 105.45.5.254
 - Upstream gateway: **none**
 - WebConfigurator: **HTTPS only**
 - GUI accessible via: `https://105.45.5.1`
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4. WAN Configuration

- WAN IP: **10.250.2.5 /29**
 - Upstream gateway: **10.250.2.1** (Neutron router)
 - NAT: Enabled
 - Block private networks: **Disabled**
 - DHCP: Disabled
 - IPv6: Disabled
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5. Initial Connectivity Validation

Upstream

- Ping: `ping 8.8.8.8`
- Test DNS via rdp-client before proceeding.

Downstream

- Ping internal hosts (e.g., 105.45.5.72).

Lateral

- Ensure reverse-path traffic flows through pfSense.
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6. Firewall Rules

WAN Rules

- Allow ICMP (Echo Request)
- Allow TCP/22 for SSH bastion
- Allow UDP/51820 for WireGuard

LAN Rules

Default allow is acceptable for lab environment.

7. SSH Bastion Configuration

Enable SSH in pfSense: - System → Advanced → Enable Secure Shell - Enable **Agent Forwarding**

Add user public keys: - System → User Manager → Users → Edit → Authorized Keys

Local .ssh/config:

```
Host pfsense
  HostName <floating-ip>
  User admin
  IdentityFile ~/.ssh/id_ed25519

Host open_int
  HostName 105.45.5.72
  User ubuntu
  ProxyJump pfsense
  IdentityFile ~/.ssh/id_ed25519
```

SSH traffic now flows:

```
ext_host → pfSense → internal hosts
```

8. NAT Considerations

NAT must remain enabled so: - Return traffic from internal-net knows how to exit - Upstream routers do not need internal subnet knowledge

If WireGuard full tunnel is deployed, additional outbound NAT rules may be required.

9. GUI Access (Temporary Method Before Bastion)

Because pfSense GUI is only accessible from LAN: 1. rdp-client temporarily attached to LAN. 2. RDP/xfce installed. 3. GUI reached at `https://105.45.5.1`.

Replaced later by WireGuard operator access.

10. Final pfSense Responsibilities Summary

- Border firewall between OpenStack and internal networks
- Routing between transit and internal subnets
- NAT for outbound internal traffic
- DHCP for internal network
- WireGuard VPN termination
- SSH bastion for operator and admin access
- Suricata host for NIDS visibility
- Syslog-ng transport for Suricata → Wazuh

pfSense forms the backbone of the VSOC environment, enforcing segmentation, enabling visibility, and acting as the central control node for secure access and monitoring.